Latent Strategies for Ad-Hoc Human–Robot Teaming in Overcooked-AI

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***Abstract*—Autonomous agents in human environments must be able to collaborate with people whose intentions are implicit and not communicated to the agent directly via language instruction. The challenge of communication is amplified in elaborate and nuanced task settings, where there are multiple pathways towards effective collaboration. Overcooked serves as a testbed for studying this form of human–robot teaming. However, the typical reactive Reinforcement Learning (RL) methodologies fall behind in terms of efficient coordination due to their limited adaptation to human strategy changes. Hence, we propose the use of strategy inference via joint, latent representations that act as a map from human behavior to the agent's appropriate response, followed by a scoring module that samples from possible strategies and generates low-level actions with a Model Predictive Path Integral (MPPI) planner, effectively bridging high-level intent with low-level actions. We benchmark our method against RL and Behavior Cloning baselines. We demonstrate that our approach effectively adapts to human preferences while achieving higher task rewards than existing methods.**

I. Introduction & Motivation

A major ingredient for successful human-robot collaboration is the robot's ability to infer a human's intent before selecting its own actions. However, while language is a powerful method to convey intent, there are multiple scenarios where intent has to be inferred from the human's behaviour. Examples of such scenarios include high-intensity situations in kitchens, as well as catastrophes like fires. Additionally, most of these scenarios have very detailed and rich task structures, involving long action sequences with many possible execution orders. The ability to correlate these implicit cues with these high-level strategies is of utmost importance to successfully coordinate and complete a task with a human partner agent.

An excellent environment to evaluate the ability of robots to perform in such scenarios is the Overcooked environment[3], which simulates a gridworld kitchen where the human and robot coordinate to prepare the most dishes in the shortest time. It offers a rich task structure via customizable recipes and enables the robot to learn collaboration solely from observed gridworld movements and actions, hence perfectly simulating the scenarios described in section I.

Most Overcooked-based human–robot teaming approaches optimize solely for task performance, leaving agents unable to interpret out-of-distribution human behavior such as novice or non-optimal play. To address this, we study joint human–robot strategy search as a foundation for efficient collaboration in rich task settings. We propose a unified policy that maps observed human actions to discrete strategies, producing interpretable agent behavior. Our data-driven approach learns joint latent representations of human and agent strategies, implicitly inferring human intent without hand-crafted HRI heuristics. A shared decoder generates candidate robot strategies, while a Model Predictive Path Integral (MPPI) planner selects the optimal strategy with performance guarantees. Our contributions can be summarized as follows:

Joint Strategy Encoder for Intent Inference: We propose a VAE-based architecture that learns joint human-robot strategy representations for intent inference in collaborative tasks.

Multi-Strategy Optimization: We use an MPPI planner with a cost function combining predicted strategy compatibility and task rewards to select optimal strategies from multiple candidates.

Evaluation Against Baselines: We evaluate against RL and BC baselines using quantitative task performance, behavioral preservation metrics, and qualitative latent space analysis.

II. Related Work

Intent Inference in Human-Robot Interaction: Intent inference is a central problem in Human–Robot Interaction (HRI), where robots reason about human goals to plan accordingly. Prior work explores predictability, legibility [6], confidence-aware planning [7], assistive autonomy [13], and intent-aware planning [25]. However, most approaches rely on task-specific heuristics and are evaluated in simple domains such as lane changes or pick-and-place. Extending intent inference to rich task environments remains challenging due to ambiguous action sequences, multiple plausible goals, and the absence of a single optimal assistive response. Moreover, without language as a communication channel, approaches relying on language-based corrections or uncertainty reduction are not applicable [14, 5, 22]. As a result, current intent inference methods are ill-suited for complex collaborative settings, making Overcooked a valuable benchmark for studying robot learning under ambiguity and nuanced coordination without language [8].

Overcooked for Human-AI Coordination: Multiple works have used the Overcooked environment to evaluate their algorithms, which include partner diversification methods[16, 4], Entropy Population based training for zero shot human-AI coordination[31] and Ad-Hoc Teaming under partial observability[23]. However, the algorithms do not factor in effective coordination with humans into their architecture design and lack interpretability for the human as they are more suited for optimizing task performance[19]. Moreover, most algorithms use Behaviour Cloning to teach RL agents to mimic human data, which does not learn the strategies the human must have used, but the general actions, which fails in out-of-distribution cases[16]. **We hypothesize that correlating low-level behaviour to these abstract strategies through joint encoded representations allows for non-invasive collaboration, while still leading to good overall task performance (H1)**.

Strategy-Level Intent Inference: Discretizing noisy human internal states into strategy clusters has been explored in HRI. While metrics such as mutual information [10] and models of noisy rationality [1] are widely used, directly leveraging them for action selection remains challenging. As a result, approaches such as information gathering over human internal states [24] and active uncertainty reduction [12] do not readily extend to rich environments like Overcooked. In high-level task settings, Nguyen et al. [20] show that sub-task decomposition improves performance, but this paradigm is insufficient for Overcooked, where real-time collaboration and strategy inference are required in addition to sub-task planning. Only a few works address strategy inference in Overcooked: Xie et al. [29] infer high-level strategies without integrating them into low-level planning, decoupling tightly coupled decision layers, while Zhao et al. [30] rely on unsupervised Hidden Markov Model-based clustering followed by strategy-specific policy learning, resulting in a cumbersome and impractical pipeline. In contrast, our strategy encoders learn correlations between human actions and agent responses without requiring a policy lookup table [30]. While self-play and cross-play are common in Overcooked, Sarkar et al. [26] propose mixed-play to induce strategy diversity, but this leads to unstable, high-variance training. **Hence, we hypothesize that jointly evaluating high-level strategies and short horizon actions in a single optimization yields higher performance than comparable baselines in collaborative human-robot tasks (H2)**. By coupling a strategy decoder with a low-level MPPI planner and scoring module, we bridge high-level strategy reasoning with low-level control, enabling a stable mapping from discrete strategies to continuous actions.

III. Problem Formulation

We formulate the problem of human-robot teaming as an ad-hoc collaboration task within the Overcooked environment. In this setting, an autonomous agent (R) must collaborate with a human partner (H) to complete a shared task without prior coordination [18]. The core challenge is that while the agent can observe the environment and human state updates, the human's internal strategy (intent) remains hidden. The agent only has access to the human's trajectory history up to the active timestep. The objective for the agent can be simplified to: 1) Observe the human's trajectory 2) Infer the human's strategy 3) Act in support of that strategy.

By strategies, we are referring to high-level task behavior with the goal of completing dishes. Dishes are completed by moving ingredients into pots, transferring the processed food onto plates to complete a dish, and then delivering that dish. For example, a human may decide to exclusively move ingredients into pots as their strategy, leaving all plating and delivering to the agent.

We mathematically formulate our problem as a Partially-Observable Markov Decision Process (POMDP), where the partial observability refers to the unknown human strategy or policy π^H. We define the problem tuple as:

$$\mathcal{M} = \langle S, A^R, A^H, \mathcal{T}, R^R, O^R, \rho_0, \gamma \rangle \tag{1}$$

Where $s \in S$ is the environment state (ingredient and item locations, human and agent locations, cooking pot status), $a^R \in A^R$ are the discrete agent actions A^R: $\{\uparrow, \downarrow, \leftarrow, \rightarrow, \otimes\}$, $a^H \in A^H$ are the discrete human actions A^H: $\{\uparrow, \downarrow, \leftarrow, \rightarrow, \otimes\}$, $\mathcal{T}$: $s \times (A^H, A^R) \rightarrow S$ is the environment transition matrix, R^R the agent rewards, $o^R \in O^R$ the observations accessible to the agent, ρ_0 the prior over Π^H, where Π^H is the unknown set of possible human policies, and γ the discount factor. Interactions with the environment ($\otimes$) include picking up or using an item. Both the agent and the human have access to S, A^R, and A^H at every timestep, though the human must process this information visually.

The ad-hoc nature of the collaboration is addressed by treating the human's underlying policy π^H as an unobservable latent variable. The agent must maintain a belief state $b_t(\pi^H)$, representing the posterior probability distribution over all possible human policies given the available history of the collaboration from the first timestep to the current timestep for the state observations $o^R_{0:t}$, and the actions $a^R_{0:t-1}$ up to but not including the current timestep:

$$b_t(\pi^H) = P(\pi^H | o^R_{0:t}, a^R_{0:t-1}) \tag{2}$$

Therefore, the agent's policy is belief-conditional. It attempts to map the inferred belief along with the current environmental and player states to an action, $\pi^R(s_t, b_t(\pi^H)) \rightarrow a^R_t$, so enabling the agent to adjust it's behavior as more information becomes available. The agent's true objective is then to find an optimal policy π^{R*} that seeks to maximize the expected cumulative reward from the current timestep t to the last timestep T, which we update at every timestep:

$$\pi^{R*} = \underset{\pi^R}{\operatorname{argmax}} \mathbb{E} \left[\sum_{k=t}^{T} \gamma^t R(s_k, a^R_k, a^H_k) \right] \tag{3}$$

IV. Proposed Approach

Solving the full POMDP as outlined in Equation (1) is computationally intractable due to the complexity of inferring intent in environments like Overcooked. Therefore, we take the latent strategy approach from **H1**. We make the assumption that the number of distinct human policies π^H is not infinite, and that these policies can be clumped into recognizable strategy clusters, as shown by [30]. Furthermore, we treat the human's trajectory τ^H as a sufficient proxy for the hidden policy π^H. A trajectory τ includes player states and actions,

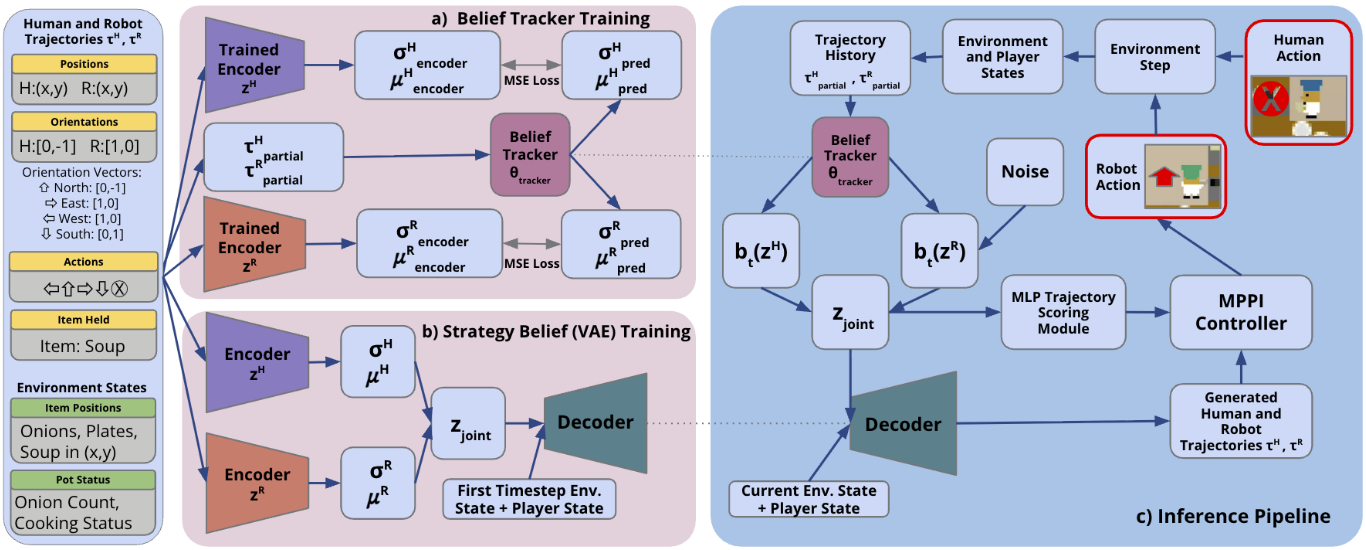

Fig. 1. Overview of the joint strategy architecture for human-robot teaming in Overcooked. **Subfigure (a)** Belief tracker training, where an LSTM predicts human and robot latent strategy encodings from partial trajectories. **Subfigure (b)** VAE-based strategy belief training, which encodes human-robot trajectories into a latent embedding z_{joint} and reconstructs future states and actions using the decoder. **Subfigure (c)** Inference pipeline, where the belief tracker and decoder generate candidate joint trajectories that are scored by an MLP+MPPI controller to select the robot action best supporting the inferred human strategy.

as depicted in Figure 1. Since $S \rightarrow A^H$ is unobservable, we use the sequence of outputs the policy produces over time (τ^H) as the indicator for intent. This assumption is supported by the concept of inverse reinforcement learning, where the underlying objective function is predicted using behavior observations [9].

Because the strategy space is structured and limited, observing τ^H provides sufficient information to identify the strategy mode that the human is currently executing, as we make the assumption that the human is not acting randomly. This allows us to avoid tracking the belief $b_t(\pi^H)$ over the infinite policy space Π^H, and instead model intent as a low-dimensional latent variable z. This enables the reduction of the POMDP to a tractable Latent-MDP. We augment the environment state s_{env} with the latent strategy z, resulting in the augmented state $\tilde{S} = (s_{env}, z)$. We define the Latent-MDP as:

$$\mathcal{M}_{latent} = \langle \tilde{S}, A^R, \tilde{\mathcal{T}}, \tilde{R}, \gamma \rangle \tag{4}$$

Where $\tilde{\mathcal{T}}(\tilde{s}_{t+1}|\tilde{s}_t, a_t)$ is the transition function learned using joint embeddings and $\tilde{R}(\tilde{s}_t, a_t)$ is the overall environment reward evaluated on a predicted joint future. $\tilde{\mathcal{T}}$ requires a joint representation of both human and robot behavior, which we create using latent space embeddings[2, 27] and is given by:

$$z_{joint} = f_\theta(\tau^H, \tau^R) \tag{5}$$

Where τ^H, τ^R are the human and agent trajectories that are being passed as input, and f_θ is the learned encoder. The training data contains 480 recorded trajectories pairs from human-human 60 second Overcooked games, the collection of which was performed by the authors of this paper. More information on data collection can be found in Appendix A.

A. Architecture

The system is divided into three primary components, as seen in Figure 1. (1) **Strategy Belief Encoding**: learns a latent representation of the collaborative behavior. (2) **Belief Tracker**: used for predicting the strategy belief encoding. (3) **Inference Pipeline**: achieves real-time control by assessing generated trajectories using an MPPI controller. The training hyper-parameters can be found in Appendix C.

(1) Strategy Belief Encoding: We use a Variational Autoencoder (VAE)[15] to learn a latent space that encodes collaborative strategies found in the training data. During training, the full trajectories are split up using a sliding window of 50 timesteps. First, two separate Long Short-Term Memory (LSTM) [11] encoders are trained on these 50 timestep trajectories and environment states s_{env}:

$$Encoder^H(\tau^H, s_{env}) \rightarrow (\mu_z^H, \sigma_z^H) \tag{6}$$

$$Encoder^R(\tau^R, s_{env}) \rightarrow (\mu_z^R, \sigma_z^R) \tag{7}$$

The corresponding latent variables z^H and z^R are then sampled from the distributions described by the resulting means (μ_z^H, μ_z^R) and standard deviations (σ_z^H, σ_z^R) via the reparameterization trick [15]. Finally, a joint strategy embedding z_{joint} is formed by concatenating the individual latent variables:

$$z_{joint} = [z_H \mid z_R] \tag{8}$$

A single decoder is used to take the joint latent embedding z_{joint} and the environment states $s_{initial}$ at the initial timestep in the data window to reconstruct the original joint trajectory:

$$Decoder(z_{joint}, s_{initial}) \rightarrow (\hat{\tau}^R, \hat{\tau}^H, \hat{s}_{env}) \tag{9}$$

Therefore, the decoder also implicitly learns a latent dynamics model of the environment. This training process is shown in Figure 1 b). In terms of loss, cross-entropy for the one-hot encoded actions, MSE loss for the environment states, and Kullback-Leibler (KL) divergence loss terms were used.

(2) Belief Tracker: The belief tracker is an LSTM network that is trained on partial trajectories of the human's behavior. These partial trajectories range from 1-50 timesteps, with randomized windows. The LSTM outputs the predicted mean and standard deviation of the human's (μ_z^H, σ_z^H) and robot's (μ_z^R, σ_z^R) latent strategy VAE encoding, using the encoders from the VAE training as the ground truth for the Mean Squared Error (MSE) loss. A belief state of both the human's strategy $b_t(z^H) \rightarrow \hat{z}_t^H$ and a belief over a potential robot strategy pairing strategy $b_t(z^R) \rightarrow \hat{z}_t^R$ are generated to give us an estimate of the expected human behavior, as well as what latent variables for the robot behaviors are commonly associated with that human strategy.

(3) Inference Pipeline: The main goal during inference is to take in the available trajectory and environment state history, generate potential future trajectories (this includes actions), and then select the robot actions with the greatest environmental reward potential. Since a full 50 timesteps are not initially available, the trajectory history is fed into the belief tracker. The belief over $b_t(z^H)$ is directly used for the joint latent embedding, while the belief over $b_t(z^R)$ is injected with noise ($\epsilon_i \sim \mathcal{N}$) to generate different trajectories:

$$z_{joint}^{(i)} = [\hat{z}_t^H \mid \hat{z}_t^R + \epsilon_i]; \quad \epsilon_i \sim \mathcal{N}(0, I) \tag{10}$$

Using this method, we set a total of 10 different z_{joint} to be created, leading to 10 generated trajectory pairings by the decoder. The MPPI controller then assesses each traj. pair:

$$J(z_{joint}^{(i)}) = \left[\sum_{k=t}^{t+H} R_{env}(\hat{s}_k, \hat{a}_k^A, \hat{a}_k^H)\right] + MLP(z_{joint}^{(i)}) \tag{11}$$

Where the sum over R_{env} is the reward for a given trajectory rolled out in the simulated environment, from the current timestep t to the final generated timestep $t+H$ where H is the length of the generation horizon. The Multi-Layer-Perceptron (MLP) component predicts the total environmental reward for a full collaboration horizon given a joint latent embedding, and is trained on our collected data. This combination of short horizon actions along with high-level strategies in a single reward function formalizes **H2**.

V. Results and Discussion

Baselines: We compare against two established baselines: (1) Proximal Policy Optimization (PPO) with Fictitious Co-Play [21], which develops strategies through self-play with diverse partners, and (2) Behavior Cloning (BC) trained on our human demonstration dataset, providing direct comparison between online optimization and imitation learning. Both are standard approaches in Overcooked-AI research [3, 30, 28].

Our evaluation uses two scenarios reflecting different interaction contexts. **Human replay scenarios** (Table I) test adaptation to fixed, pre-recorded human trajectories that replay without responding to our agent, placing complete adaptation burden on our method. We evaluate three strategies (onion-only, plate-only, plate+onion) from both left and right positions, representing cases where humans are unfamiliar with the system, cognitively loaded, or unwilling to modify their behavior. **BC proxy scenarios** (Table II) use the BC agent as an adaptive human proxy exhibiting human-like behavior with capacity to respond to partner actions, simulating responsive partners constrained to human-like patterns. Together, these scenarios evaluate performance in worst-case (non-adaptive) and realistic (adaptive partner) conditions. Performance is measured by cumulative reward over 200 episodes per condition.

A. *Model Evaluation*

Non-Adaptive Human Replay Results: Table I reveals distinct performance patterns. The RL baseline achieves highest performance in two scenarios: onion-only/human-right (96.2) and plate-only/human-left (103.3), but completely fails in complementary configurations (zero reward). This asymmetry stems from RL training collapse: despite alternating between six partners during self-play, all agents converged to a single rigid pattern (right handles onions, left handles plates), producing an agent optimized for one pairing but unable to adapt when violated.

Our method achieves best performance across all other strategy-position combinations. Only in the two scenarios matching RL's collapsed strategy does RL exceed our performance, where we still achieve 80% of RL's reward while maintaining strong performance elsewhere. We substantially outperform BC across all scenarios (1.2-2.2$\times$ improvement), demonstrating that online strategy optimization provides significant advantages over imitation learning alone.

Adaptive BC Proxy Results. Table II shows our method achieves highest reward in both positions (90.6 and 86.3), outperforming BC-BC by 72.5% and 64.6%, and BC-RL by 47.7% and 23.3%. The substantial improvement over BC-BC—two agents trained on identical demonstrations—indicates that online MPPI optimization discovers more effective collaboration patterns than those in the average human demonstrations.

Adaptation vs. Specialization. These results highlight a fundamental trade-off. RL specialization yields excellent performance within one pattern but fails when violated. BC maintains consistent modest performance but cannot achieve optimal rewards. Our approach balances these extremes through learned representations with online optimization, achieving strong performance across all partner behaviors.

Our method's consistent performance across diverse strategies addresses a critical deployment challenge: humans naturally vary their behavior based on experience, preferences, and cognitive state. Systems that fail when humans deviate from expected patterns—as RL demonstrates—pose usability and safety concerns. Our approach accommodates behavioral diversity without requiring users to learn specific interaction

TABLE I

TASK PERFORMANCE WITH NON-ADAPTIVE HUMAN REPLAY PARTNERS EXECUTING FIXED STRATEGIES. MEAN $\pm$ STANDARD DEVIATION OVER 200 EPISODES. BOLD INDICATES BEST PERFORMANCE.

Agent Pairing	Human Strategy		
	Onion Only *Right / Left*	Plate Only *Right / Left*	Plate & Onion *Right / Left*
Human-RL	$\mathbf{96.2 \pm 53.9}$ / 0.0 ± 0.0	0.0 ± 0.0 / $\mathbf{103.3 \pm 67.0}$	40.5 ± 49.8 / 20.2 ± 32.1
Human-BC	49.8 ± 44.5 / 18.7 ± 20.7	24.3 ± 19.3 / 39.3 ± 40.2	45.9 ± 32.6 / 33.0 ± 30.9
Human-MPPI (Ours)	83.4 ± 46.5 / $\mathbf{41.5 \pm 33.7}$	$\mathbf{30.5 \pm 18.6}$ / 80.3 ± 53.8	$\mathbf{47.2 \pm 34.0}$ / $\mathbf{35.7 \pm 29.8}$

TABLE II

TASK PERFORMANCE WITH ADAPTIVE BC PROXY PARTNERS. MEAN $\pm$ STANDARD DEVIATION OVER 200 EPISODES.

Agent Pairing	Environment Reward *BC Right / Left*
BC-RL	61.3 ± 51.4 / 70.0 ± 78.7
BC-BC	52.4 ± 37.3 / 52.4 ± 37.3
BC-MPPI (Ours)	$\mathbf{90.6 \pm 53.6}$ / $\mathbf{86.3 \pm 73.2}$

TABLE III

BEHAVIORAL ADAPTATION REQUIRED FROM BC AGENT. MSE QUANTIFIES DEVIATION FROM BC-BC BASELINE; LOWER VALUES INDICATE LESS ADAPTATION. RESULTS FOR LEFT AND RIGHT POSITIONS.

Agent Pairing	Adaptation MSE *(BC Left / BC Right)*
BC-RL	$\mathbf{0.0001}$ / 0.0064
BC-MPPI (Ours)	0.0032 / $\mathbf{0.0026}$

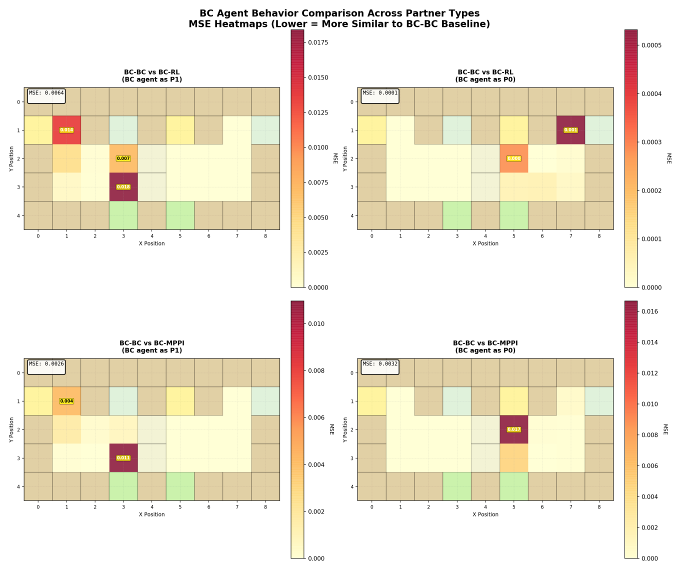

Fig. 2. BC agent spatial occupancy across partners. Top: BC-BC baseline. Bottom: BC with RL and MPPI (left/right). Lower deviation indicates better behavior preservation.

patterns, functioning effectively in both non-adaptive and adaptive scenarios across novice users, high-workload situations, and established workflows.

Human Behavior Preservation: Minimizing disruption to human collaborators' natural behavior is critical. Forcing humans to alter strategies can increase cognitive load, reduce satisfaction, and introduce safety concerns. We evaluate our approach's impact by comparing spatial occupancy patterns of a BC agent (human proxy) paired with different partners.

We compute spatial occupancy heatmaps over the environment grid, recording agent occupancy frequency per cell. We quantify behavioral change using MSE between BC-BC baseline heatmaps and BC-MPPI/BC-RL pairings. Higher MSE indicates greater forced deviation. The asymmetric layout requires separate left/right evaluation (Table III).

Figure 2 visualizes spatial occupancy across pairings. On the left, RL achieves lowest MSE (0.0001). On the right, our MPPI achieves lower MSE (0.0026 vs. 0.0064), requiring less behavioral change. This asymmetry reflects RL's training collapse to right-optimized strategies, forcing greater BC partner compensation on the left.

Absolute MSE values are small (10^{-3} to 10^{-4}), indicating limited BC flexibility regardless of partner. This stems from BC's pure imitation training producing rigid execution. Actual humans would likely show greater adaptation. Despite this BC limitation, our method achieves competitive performance (Table II) while requiring minimal partner behavioral change.

Preserving partner behavior while maintaining performance demonstrates a key framework advantage. By encoding diverse strategies in latent space and optimizing online, our approach discovers complementary actions accommodating natural partner behavior rather than imposing fixed coordination. This contrasts with end-to-end policies (BC, RL) that assume specific partner behaviors from training and struggle when partners deviate. RL's position-dependent performance exemplifies this: training converged to a narrow pattern effective in one configuration but failing to generalize.

Systems accommodating diverse behaviors without requiring user adaptation achieve better adoption, particularly where users have established workflows or high cognitive load. Our results suggest combining learned representations with online optimization enables flexible systems that adapt to humans rather than forcing humans to adapt.

Latent Space Clustering: Our method relies on the learned latent space structure encoding strategies efficiently. Figure 3 visualizes the latent space via PCA: the left panel shows points colored by original strategy labels; the right shows K-means clustering.

The Normalized Mutual Information (NMI) between orig-

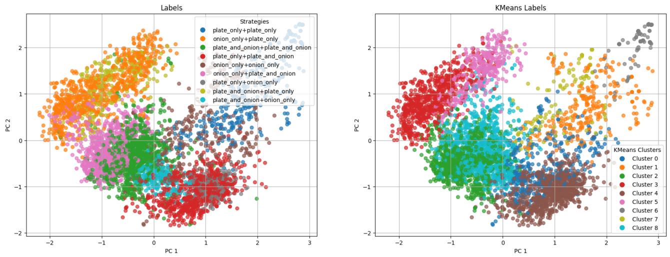

Fig. 3. Latent space visualization using PCA. Left: Points colored by original strategy labels. Right: K-means clustering. NMI of 0.5880 indicates the latent space groups strategies by functional similarity rather than semantic labels.

inal labels and K-means is 0.5880 ($\sim 60\%$ agreement). This partial alignment reflects a key strength: the VAE encodes *functional similarity* rather than semantic labels. Many labeled strategies are functionally identical—for example, left-side agents permitted to use onions and plates often choose plates only for efficiency. Consequently, "plate+onion" trajectories may be functionally equivalent to "plate-only" despite different labels. The VAE automatically merges functionally similar strategies into shared regions, discovering true strategic structure rather than learning the arbitrary semantic categorization.

We observe clear clustering: each strategy concentrates in a distinct region with identifiable mean and variance. Overlaps occur only between highly similar strategies, indicating the VAE learned a space where proximity corresponds to strategic similarity—essential for effective trajectory generation, as local perturbations produce local strategy variations.

Furthermore, smooth transitions exist between clusters: plate-only interpolates smoothly into plate+onion, which interpolates into onion-only. When the belief predictor is uncertain, these smooth transitions ensure samples between cluster means produce meaningful trajectories. Neighboring points correspond to functionally similar strategies, enhancing robustness.

The resulting latent space—with clear clustering, smooth transitions, and strategic similarity alignment—naturally facilitates MPPI trajectory generation for efficient partner adaptation.

VI. Conclusion

We formulated ad-hoc human-robot teaming as a Latent-MDP, where the human's policy is a low-dimensional latent variable inferred from trajectories. Our architecture implements this using a VAE that learns joint human-robot strategy embeddings, a belief tracker over these embeddings, and an MPPI controller that evaluates generated trajectories. Our results support **H1**: correlating low-level behavior with abstract joint strategy embeddings enables non-invasive collaboration without sacrificing task performance. In both human replay and BC proxy settings, the MPPI+VAE agent attains competitive rewards relative to BC and RL baselines, even with non-adaptive partners. Heatmaps of human behavior disruptions show that our method can require less deviation from a partner's baseline behavior than other models, indicating adaptation to human preferences rather than forcing strategy changes. Finally, joint optimization over short-horizon rollouts and latent strategies **(H2)** yields consistently high rewards, with only limited cases where a specialized RL policy performs better, suggesting our method is a promising approach for flexible, non-invasive ad-hoc teaming in rich tasks.

A. *Limitations and Future Work*

This work is limited to a single asymmetric Overcooked layout, a fixed onion-soup recipe, and a relatively small set of human–human demonstrations (480 trajectories), so it is unclear how well the learned latent strategies and MPPI controller would generalize to other tasks or environments. Additionally, our work is limited by a lack of direct human trials with our method. To conclusively test our method, a user study where participants rate our method against other baselines would be required (see Appendix B for an outline).

A more fundamental limitation is that our latent-strategy framework currently lacks theoretical guarantees to find a possible collaborative strategy for our agent if one exists. Even with strong performance, we cannot characterize how belief updates behave under distribution shift, or how close the MPPI controller is to an optimal joint policy. Future work could therefore explore trade-offs on robustness, sample efficiency, and interpretability of the learned latent strategies to improve our probability of finding an effective collaborative strategy.

VII. Contributions of Each Team Member

Mehul: I have contributed towards the ideation steps and formulating the methodology. Additionally, I collected data for around 288 games for training, and implemented a modified BC pipeline for training the agent on our data, alongside the heatmaps for evaluation. In this report, I have written the abstract, introduction and the related works sections.

Adrian: I coded the architecture pipeline including VAE training, belief tracker training, data-loading, PCA analysis, and inference script. This pipeline was iterated over in collaboration with Nate. I created the final data collection protocols and collected 288 training episodes. I assisted Nate with conceptualizing evaluation criteria. I wrote the Problem Formulation, Proposed Approach, and Conclusion sections.

Nate: I proposed the our methodology. I designed, implemented, and trained the VAE, Belief Prediction, and Scoring modules (with Adrian), implemented and designed the MPPI cost function and planning loop for our method and modified the Overcooked environment for integration. I collected 192 training episodes and trained the RL baselines. I conducted all experiments and produced all plots and tables. I wrote the Results and Conclusion sections and revised the entire manuscript.

Juee: I contributed towards the ideation steps and the methodology. I did the analysis of previous work in the beginning for the midterm report and collected 192 training episodes. I contributed to the MPPI structure and designed the human study. In this paper, I wrote the appendix.

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Appendix A
Data Collection

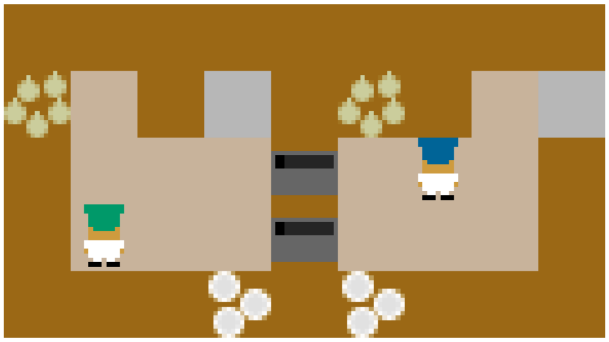

Fig. 4. Asymmetric Advantages Layout

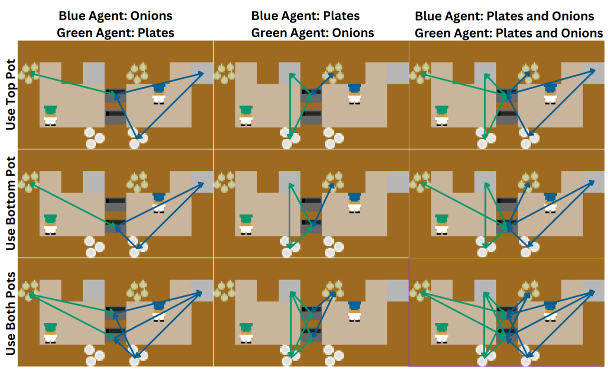

Fig. 5. Asymmetric Advantages Strategies

Our chosen layout, seen in Figure 4 has two separated rooms where players are confined to their individual rooms, but both players share the same pots. The two players have different distances to environment objects such as plates, delivery counters, and onions, meaning that there are many divisions of how players may choose to build and deliver recipes, even if not all combinations are optimal. The player decisions focus on task divisions, such as which player gathers the onions or which agent delivers the recipes, and navigation trade-offs such as how far each player is from objects in the environment. This design minimizes collision-related costs and eliminates the need of low-level motion planning, allowing strategic variation to be the main driver of observed behavior.

The recipe used across all data collection trials was onion soup, chosen for its simplicity. Each data-collection episode lasted 60 seconds. To potentially elicit distinct and interpretable strategic modes, we used a leader–follower pairing where one human player was assigned a leader role, while the other acted as a follower. Both players were assigned predefined strategies to execute during the game. The strategies that the humans were asked to play are given in Table IV, and shown visually in Figure 5.

For each human-human pairing, this experimental setup was conducted four times, where each human was both the leader and follower for each respective side. There were a total of 5 human pairings, and the total number of trajectories collected was 480.

TABLE IV
Experimental Setup for Custom Overcooked Data Collection

Player 1 Strategy	**Player 2 Strategy**	**# of Plays**
Onion Only	Onion Only	2
Plate Only	Plate Only	2
Plate Only	Onion Only	5
Plate+Onion	Plate+Onion	5
Onion Only	Plate+Onion	5
Plate Only	Plate+Onion	5

Appendix B
Proposed User Study

The question we are trying to address with this study is that does strategy encoding create a better agent for human–robot game play and interaction. We hypothesize that the method we used to train our agent will:

- act as a better partner to human players in the Overcooked-AI environment in terms of *comfort* and *trust*,
- generate a higher environmental reward,
- adapt to the human's strategy by performing complementary assisting tasks.

A. Experimental Factors

To design an experiment, it is important to identify the variables for the study. The independent variables determine the combinations and the number of game plays. We identified three main independent variables we wanted to test.

- Method: This category of variables lists all the methods which can be used as the agent for game-play. It consists of the existing baselines and our proposed method.
  - Multi-Agent Reinforcement Learning (MARL)
  - Behavior Cloning (BC)
  - Our strategy-encoded agent
- Playing Side: The playing side determines which part of the layout is the human participant playing from. This is important to consider because the layours are not symmetrical and hence affect playing strategies and the environmental reward. The human participant may play as either:
  - Player 1: Left side of the layout
  - Player 2: Right side of the layout
- Human Strategy Conditions: To test the algorithms, it is important that we test them when the human plays adversarially to the goal of the game, and collaboration is essential to complete the tasks. Single item strategies like onion only or plate only are examples of this and require collaboration to deliver soups. We also want to test algorithms in normal conditions, wherein humans can play however they want and the agent is assistive.

– Onion Only: Human uses only onions.
– Plate Only: Human uses only plates.
– Free Game play: Human may freely use both plates and onions to maximize reward.

The control variables for the experiment determine the experiment setup and the protocol. We decided to use the layout identical to that used during data collection and the same recipe - onion soup. The game duration is also fixed at 1 minute. The dependent variables are the ones which help us analyze the results and validate the hypothesis. They are the environmental reward score and the data regarding trust and comfort obtained from the Likert scale survey.

B. Study Design

The structure of the study should be within-subjects, where each participant performs all experimental conditions. The total number of gameplays per participant is determined by the factorial structure of the independent variables:

$$3 \text{ methods} \times 2 \text{ playing sides} \times 3 \text{ strategies} = 18 \text{ gameplays}$$

TABLE V
FACTORIAL COMBINATION OF METHODS, STRATEGIES, AND HUMAN PLAYER ROLES

Strategy / Method	**MARL**	**Behavior Cloning**	**Our Agent**
Onion Only	P1, P2	P1, P2	P1, P2
Plate Only	P1, P2	P1, P2	P1, P2
Free Play	P1, P2	P1, P2	P1, P2

The study is planned for 30 minutes per participant. The breakdown of the time is listed below:

- 6 minutes: Introduction to the study
- 8 minutes: Method A game-play + questionnaire
- 8 minutes: Method B game-play + questionnaire
- 8 minutes: Method C game-play + questionnaire

To ensure the validity and reliability of the results, we would have to give names as A,B and C to the methods instead of displaying their actual agent algorithm names to the participants. We will randomize the order of Method A, B, and C is randomized for each participant.

C. Questionnaire

The survey questionnaire will be asked to filled after every method is played. We will collect the name of the method, the environmental rewards for each game-play and ratings of the Likert scale [17] of 1 to 5, with 1 being ”least agree with” and 5 being ”strongly agree with” for the following questions:

- The agent understood what I wanted to do
- The agent performed as a team and we could deliver more than what I would have individually been able to do
- I trust the agent and felt comfortable playing

APPENDIX C
MODEL TRAINING HYPER-PARAMETERS

TABLE VI
ARCHITECTURE AND TRAINING HYPERPARAMETERS FOR THE BELIEF TRACKER AND VAE.

Module	Hyperparameter	Value
Belief Tracker	Input Dimension	16
	Latent Dimension	32
	Hidden Dimension	256
	Max Sequence Length	50
VAE	Input Dimension	16
	Latent Dimension	32
	Hidden Dimension	256
	Sequence Length	50
Strategy Scoring Module	Input Dimension	64
	Hidden Dimension	256
	Sequence Length	50
Training	VAE Epochs	300
	Tracker Epochs	150
	Batch Size	64
	Learning Rate η	10^{-4}
	State Dimension $\mathcal{S}$	76
	Action Dimension $\mathcal{A}$	6